Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 1 / 方法 1 [GE06-M01]

Tangency has an orientation: external circles have center distance $r_1 + r_2$, internally tangent circles have $|r_1 - r_2|$. Identify the correct center ray, side of each tangent and radius sign before eliminating variables.

相切具有方向：外切圆心距为 $r_1 + r_2$，内切为 $|r_1 - r_2|$。消元之前先确定圆心射线、切线两侧及半径正性。

Center-distance equations need ray and containment checks

圆心距离方程需要射线与容纳检验

Procedure

1. Use symmetry or angle bisectors to locate the unknown center with as few variables as possible.

2. Write each tangency as a radius sum or difference, keeping absolute values until the center order is known.

3. Solve and restore all unsquared signs, positive radii, segment contact positions and containment inequalities.

通用步骤

1. 利用对称或角平分线，以尽可能少的变量定位未知圆心。

2. 将各相切写为半径和或差；圆心次序未明时保留绝对值。

3. 求解后补查未平方符号、正半径、线段切点位置及容纳不等式。

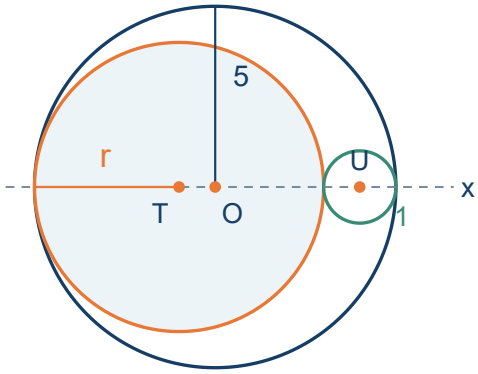
> Example. A radius-5 circle is centered at $(0, 0)$ and contains a radius-1 circle centered at $(4, 0)$. A third circle has center on the negative x axis, is internally tangent to the large circle and externally tangent to the unit circle. Find its radius.

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 1 / 方法 1 [GE06-M01] (continued / 续)

例题：圆心 $(0,0)$、半径 5 的圆内有圆心 $(4,0)$、半径 1 的圆。第三圆圆心在负 x 轴，与大圆内切、与单位圆外切，求其半径。



O = (0,0) U = (4,0) T is on −x / T 在负半轴

Worked solution

1. Write its center as $(-t,0)$ with $t>0$ and its radius as $r>0$. Internal tangency gives $t+r=5$.

2. External tangency gives $4+t=r+1$. Together these yield $t=1, r=4$.

3. The circle is contained in the large one because center distance plus radius is $1+4=5$; its center distance to the unit circle is five, equal to the radius sum.

例题解析

1. 设圆心为 $(-t,0)$，其中 $t>0$，半径 $r>0$。内切给出 $t+r=5$。

2. 外切给出 $4+t=r+1$，联立得 $t=1, r=4$。

3. 圆心距加半径为 $1+4=5$，确在大圆内；到单位圆的圆心距为五，等于半径和。

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 1 / 方法 1 [GE06-M01] (continued / 续)

A squared distance equation cannot distinguish internal from external tangency by itself.

仅凭距离平方方程无法区分内切与外切。

If a center lies on a vertex angle bisector at distance d from the vertex, its radius is $d\sin(A/2)$. When the circumcenter lies on the same line, preserve both branches of $|R-d|=R-r$.

In a symmetric isosceles triangle, the incenter and excenters have explicit axial coordinates. Internal tangency may require the new circle to contain an excircle rather than lie inside it; decide from center distance before choosing the radius difference.

圆心在顶角平分线上，距顶点为 d 时，半径为 $d\sin(A/2)$。若外心也在此线上，须保留 $|R-d|=R-r$ 的两个分支。

等腰三角形的内心、旁心可取明确轴向坐标。内切时，新圆可能包含旁切圆，而非位于旁切圆内；应先比较圆心距再选半径差。

Practice / 配套练习: AIME-2004-II-Q12 (WB p. 342), AIME-2008-II-Q11 (WB p. 344), AIME-2009-II-Q05 (WB p. 330), AIME-2014-II-Q08 (WB p. 335), AIME-2019-I-Q11 (WB p. 346), AIME-2022-I-Q08 (WB p. 336), AIME-2025-II-Q06 (WB p. 339)

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 2 / 方法 2 [GE06-M02]

Equal-circle arrays are polygons of centers

等圆排列转化为圆心多边形

Procedure

1. Replace each tangent circle by its center, with neighbor distance twice the radius.

2. For staggered rows use equilateral-center triangles; for a ring use a regular-center polygon.

3. Add radius margins to obtain enclosing dimensions, then subtract disk areas only after verifying disjoint interiors.

通用步骤

1. 将相切圆换成圆心，相邻圆心距为半径两倍。

2. 交错行使用等边圆心三角形，环形排列使用正圆心多边形。

3. 求外框尺寸时加入半径边距；扣圆盘面积前须确认内部互不重叠。

> Example. Six unit circles surround a central unit circle, all neighbors tangent. Find the radius of the smallest concentric enclosing circle.

> 例题：六个单位圆围绕一个中心单位圆，相邻圆均相切，求同心最小外包圆半径。

Worked solution

1. Each outer center is distance two from the central center.

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 2 / 方法 2 [GE06-M02] (continued / 续)

2. Neighboring outer centers also have separation two, so they form a regular hexagon of circumradius two.

3. The farthest point of each outer disk is one more unit away. The enclosing radius is 3, and it is attained on each outward radial line.

例题解析

1. 每个外圆圆心距中心圆心为二。

2. 相邻外圆心间距也为二，所以形成外接圆半径二的正六边形。

3. 每个外圆盘最远点再向外一单位，故外包半径为 3，在各向外半径上取到。

In a regular n-circle ring, $2r = 2d\sin(\pi/n)$ relates circle radius r and center-ring radius d.

正 n 圆环中，圆半径 r 与圆心环半径 d 满足 $2r = 2d\sin(\pi/n)$。

Practice / 配套练习: AIME-2002-I-Q02 (WB p. 328), AIME-2005-I-Q01 (WB p. 329)

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 3 / 方法 3 [GE06-M03]

Homothety controls common tangents and opposite circle chords

位似控制公切线与相反方向圆弦

Procedure

1. For unequal radii, common external tangents meet at the external similarity center, dividing center distances in the radius ratio. Equal-radius external tangents are parallel and have no finite similarity center.

2. Common internal tangents pass through the internal similarity center; include its directed location on the center segment.

3. At an external circle-contact point, a negative homothety maps a secant chord to the other circle; use signed distances when areas are compared.

通用步骤

1. 半径不等时，两条公外切线交于外位似中心，到两圆心距离比等于半径比；半径相等时公外切线平行，不存在有限外位似中心。

2. 公内切线经过内位似中心，应保留其在圆心线段上的有向位置。

3. 两圆外切点处，负位似将一圆割弦映至另一圆；比较面积时须使用有向距离。

> Example. Externally tangent circles have radii $1, 4$. Find the distance from the smaller center to the intersection of their common external tangents.

> 例题：两外切圆半径为 $1, 4$，求小圆圆心到两条公外切线交点的距离。

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 3 / 方法 3 [GE06-M03] (continued / 续)

Worked solution

1. The centers are five units apart. The external homothety center lies beyond the smaller circle.

2. If its distance to the smaller center is d, its distance to the larger center is $d+5$, and $(d+5)/d=4$.

3. Thus $d=5/3$. The tangent half-angle satisfies $\sin\theta=1/d=3/5$, consistent for both circles.

例题解析

1. 两圆心距为五，外位似中心在小圆外侧。

2. 设到小圆心距离为 d，则到大圆心为 $d+5$，比例给出 $(d+5)/d=4$。

3. 所以 $d=5/3$；切线半角满足 $\sin\theta=1/d=3/5$，对两圆均一致。

An external-contact homothety has negative scale, so corresponding chord points lie on opposite rays.

以外切点为中心的位似比为负，对应弦点位于相反射线。

If chord lengths from the contact point scale by R/r, equality of two triangle areas forces the opposite inverse ratio between their vertex-to-secant distances. A same-side condition determines whether those distances have the same signed ratio.

若从切点量起的弦长按 R/r 缩放，两个三角形面积相等便要求相应顶点到割线距离满足逆比例。同侧条件决定其有向比例是否同号。

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 3 / 方法 3 [GE06-M03] (continued / 续)

Practice / 配套练习: AIME-2006-II-Q09 (WB p. 333), AIME-2015-II-Q15 (WB p. 345), AIME-2022-II-Q07 (WB p. 337)

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 4 / 方法 4 [GE06-M04]

Half-angle tangent lengths solve tangential polygons and circle chains

半角切线长解外切多边形与圆链

Procedure

1. Equal tangents give the two side portions belonging to each polygon vertex.

2. For tangent length t and inradius r, write $\tan(A/2) = r/t$; the half-angle sum is $(n-2)\pi/2$.

3. Use tangent identities with branch checks; the sum of positive arctangents increases strictly with r.

4. For a row of equal circles along a triangle base, the end-center offsets are $r\cot(B/2), r\cot(C/2)$ and neighbor gaps are $2r$.

通用步骤

1. 由同点等切线长求各多边形顶点所占两段。

2. 切线长为 t、内切圆半径为 r 时，$\tan(A/2) = r/t$，半角总和为 $(n-2)\pi/2$。

3. 使用正切恒等式并检验分支；正反正切之和随 r 严格递增。

4. 三角形底边圆链的两端圆心偏移为 $r\cot(B/2), r\cot(C/2)$，相邻圆心间距为 $2r$。

> Example. A triangle has side lengths $8, 10, 12$. Use half-angle tangent lengths to find its inradius.

> 例题：三角形三边为 $8, 10, 12$，用半角切线长求内切圆半径。

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 4 / 方法 4 [GE06-M04] (continued / 续)

Worked solution

1. The semiperimeter is fifteen, so the three vertex tangent lengths are $3, 5, 7$.

2. The three half-angles sum to $\pi/2$. The tangent-sum denominator therefore vanishes: $1 - r^2(1/15 + 1/21 + 1/35) = 0$.

3. The parenthesis is $1/7$, so $r = \sqrt{7}$. The positive half-angle sum cannot be $3\pi/2$, since each half-angle is strictly below $\pi/2$.

例题解析

1. 半周长为十五，三个顶点切线长为 $3, 5, 7$。

2. 三个半角和为 $\pi/2$，因此正切和公式分母为零：$1 - r^2(1/15 + 1/21 + 1/35) = 0$。

3. 括号等于 $1/7$，故 $r = \sqrt{7}$。每个半角严格小于 $\pi/2$，所以正半角和不可能为 $3\pi/2$。

A polynomial produced by a tangent identity can retain a wrong angle-sum branch.

正切恒等式产生的多项式可能保留错误的角度和分支。

An N-circle row of radius r along base a satisfies $a = r[\cot(B/2) + \cot(C/2)] + 2r(N - 1)$. The incircle corresponds to $N = 1$. For an odd-sided tangential polygon, the linear side equations uniquely recover all vertex tangent lengths.

底边长为 a，半径 r 的 N 圆链满足 $a = r[\cot(B/2) + \cot(C/2)] + 2r(N - 1)$；内切圆对应 $N = 1$。奇数边外切多边形的边长线性方程可唯一确定全部顶点切线长。

Practice / 配套练习: AIME-2000-II-Q10 (WB p. 331), AIME-2002-II-Q14 (WB p. 340), AIME-2007-I-Q09 (WB p. 334), AIME-2020-II-Q13 (WB p. 347), AIME-2024-I-Q08 (WB p. 338)

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 5 / 方法 5 [GE06-M05]

Eliminate an unknown tangent radius to obtain an ellipse

消去未知切圆半径得到椭圆

Procedure

1. For one internal and one external tangency write $PF_1 = R - r, PF_2 = s + r$.

2. The radius cancels to $PF_1 + PF_2 = R + s$, identifying the center ellipse.

3. Check $r = R - PF_1 > 0$ and then impose a line-intersection discriminant if slope or direction is optimized.

通用步骤

1. 一内切一外切时，写为 $PF_1 = R - r, PF_2 = s + r$。

2. 消去半径得 $PF_1 + PF_2 = R + s$，即圆心椭圆。

3. 检验 $r = R - PF_1 > 0$；若优化斜率或方向，再用直线与椭圆的判别式。

> Example. Circles centered at $(-2, 0), (2, 0)$ have radii $7, 1$. Describe centers of circles internally tangent to the first and externally tangent to the second.

> 例题：两圆圆心为 $(-2, 0), (2, 0)$，半径为 $7, 1$，求与前者内切、与后者外切的圆之圆心轨迹。

Worked solution

1. The center distances sum to eight, so the ellipse has $a = 4, c = 2, b^2 = 12$.

2. Its equation is $x^2/16 + y^2/12 = 1$. Distance to the first focus ranges from two to six.

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 5 / 方法 5 [GE06-M05] (continued / 续)

3. Thus $r = 7 - PF_1$ ranges from one to five and is always positive; every ellipse point gives the required internal/external tangencies.

例题解析

1. 圆心到两焦点距离和为八，所以 $a = 4, c = 2, b^2 = 12$。

2. 椭圆方程为 $x^2/16 + y^2/12 = 1$，到第一焦点距离介于二与六。

3. 因此 $r = 7 - PF_1$ 介于一与五且始终正，椭圆各点都满足所需内切与外切。

The ellipse equation is necessary; the recovered radius makes it sufficient.

椭圆方程只先给出必要性，恢复的半径条件才保证充分性。

Practice / 配套练习: AIME-2005-II-Q15 (WB p. 419)

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 6 / 方法 6 [GE06-M06]

Power differences measure signed distance from the radical axis

圆幂差度量到根轴的有向距离

Procedure

1. Expand the powers of a point in both circles; squared point coordinates cancel.

2. The resulting affine function vanishes on the common chord line and has gradient magnitude twice the center distance.

3. Use a chord angle or center-line ratio to express the point's distance from that line, retaining its side.

通用步骤

1. 展开一点对两圆的点幂，点坐标平方项会消去。

2. 所得仿射函数在公共弦直线上为零，其梯度长度是圆心距的两倍。

3. 用弦所对角或圆心线比例表示到该直线的距离，并保留所在侧。

> Example. Two intersecting circles have radii $5, 3$. A radius-2 circle is externally tangent to both, and its center is distance one from their radical axis. Find the two original centers' distance.

> 例题：两相交圆半径为 $5, 3$，另一个半径 2 的圆与两者外切，其圆心到原两圆根轴距离为一。求原两圆心距。

Worked solution

1. The new center has powers $(5+2)^2 - 5^2 = 24$ and $(3+2)^2 - 3^2 = 16$.

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 6 / 方法 6 [GE06-M06] (continued / 续)

2. Their difference is eight. If original center distance is d, its magnitude is also $2d \cdot 1$.

3. Hence $d = 4$, consistent with the intersection condition $|5 - 3| < d < 5 + 3$.

例题解析

1. 新圆心的两圆幂分别为 $(5+2)^2 - 5^2 = 24$ 与 $(3+2)^2 - 3^2 = 16$。

2. 差为八；若原圆心距为 d，幂差绝对值又为 $2d \cdot 1$。

3. 故 $d = 4$，并满足相交条件 $|5 - 3| < d < 5 + 3$。

Use a signed affine distance when comparing points on opposite sides of the chord line.

比较公共弦两侧的点时，应使用有向仿射距离。

Along the line of the two centers, signed distance to the radical axis is affine. A prescribed division point therefore has the same weighted combination of the two center distances; convert its half-chord length by $\sqrt{R^2 - d^2}$.

沿两圆心直线，到根轴的有向距离为仿射函数。按比例分点的距离是两圆心距离的同权线性组合，再由 $\sqrt{R^2 - d^2}$ 求半弦长。

Practice / 配套练习: AIME-2005-II-Q08 (WB p. 332), AIME-2021-I-Q13 (WB p. 348)

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 7 / 方法 7 [GE06-M07]

Offset corner circles are an incenter homothety

内移角圆组成以内心为中心的位似

Procedure

1. A circle of radius x tangent to a vertex's two sides has its center on the vertex-to-incenter line.

2. For corner circles inside the triangle with $0 < x < r$, the three centers are images of the vertices under scale $(r - x)/r$ about the incenter, where r is the triangle inradius.

3. A fourth equal circle tangent to all three has its center at their circumcenter; impose circumradius $2x$ and verify the fourth disk stays inside.

通用步骤

1. 半径 x 的圆与某顶点两邻边相切时，圆心在顶点到内心的直线上。

2. 若三个角圆均在三角形内且 $0 < x < r$，圆心是三顶点以内心为中心、比例 $(r - x)/r$ 的位似像，其中 r 为原内切圆半径。

3. 第四等圆与三圆均外切时，其圆心为该圆心三角形外心，故外接半径为 $2x$；最后核实第四圆也在内部。

> Example. In an equilateral triangle of inradius one, three equal circles inside the triangle are each tangent to the two sides at a different vertex. A fourth circle of the same radius lies inside and is externally tangent to all three. Find the common radius.

> 例题：内切圆半径为一的等边三角形内，有三个等圆分别与三个顶点的两邻边相切。第四个同半径圆也在三角形内，并与前三圆均外切，求公共半径。

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 7 / 方法 7 [GE06-M07] (continued / 续)

Worked solution

1. The original circumradius is two. Corner-center homothety gives circumradius $2(1-x)$.

2. Tangency to the fourth equal circle requires $2(1-x)=2x$, hence $x=1/2$.

3. The fourth center is the triangle center, distance one from every side, so its radius-$1/2$ disk lies inside; the corner circles also fit.

例题解析

1. 原外接圆半径为二，角圆心位似给出外接半径 $2(1-x)$。

2. 第四等圆外切要求 $2(1-x)=2x$，所以 $x=1/2$。

3. 第四圆心为原中心，距三边均一，故半径 $1/2$ 的圆在内部；三个角圆也可容纳。

Equal center distances alone do not prove that the fourth disk fits inside the triangle.

到三个圆心等距本身，并不能证明第四圆盘位于三角形内。

Practice / 配套练习: AIME-2007-II-Q15 (WB p. 343)

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 8 / 方法 8 [GE06-M08]

Rolling and nested tangency use radius sums or differences

滚动与嵌套相切使用半径和或差

Procedure

1. For a circle rolling externally around a fixed circle, the moving center follows radius $R + r$.

2. For fixed-circle radius R, moving radius r, ground-frame spin ϕ, and center sweep θ, no slip gives $r\,d\phi = (R + r)\,d\theta$.

3. For nested internal tangency, center displacement has length $R - r$; if radii shrink to zero, the nested disks have at most one common point.

通用步骤

1. 圆绕固定圆从外侧滚动时，动圆心轨迹半径为 $R + r$。

2. 固定圆半径为 R、动圆半径为 r、对地自转角为 ϕ、圆心公转角为 θ 时，无滑动给出 $r\,d\phi = (R + r)\,d\theta$。

3. 嵌套内切时圆心位移长为 $R - r$；若半径趋零，嵌套圆盘至多有一个公共点。

> Example. A radius-2 circle rolls without slipping around a fixed radius-1 circle. During a 90° sweep of the center, how much does the moving circle spin?

> 例题：半径 2 的圆沿半径 1 的固定圆从外侧无滑动滚动，圆心转过 90° 时动圆自转多少度?

Worked solution

1. The moving center follows a circle of radius three, so its path length is $3\pi/2$.

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 8 / 方法 8 [GE06-M08] (continued / 续)

2. No slip makes this length equal to moving radius two times spin angle in radians.

3. The spin is $3\pi/4 = 135^\circ$.

例题解析

1. 动圆心轨迹半径为三，路程为 $3\pi/2$。

2. 无滑动使该路程等于动圆半径二乘以弧度自转角。

3. 自转角为 $3\pi/4 = 135^\circ$。

Center revolution and circle spin are different angles.

圆心公转角与圆本身自转角不同。

For a nested sequence with radii r^n and successive radial directions rotating by a fixed unit complex factor u, center increments form a geometric series proportional to $(ru)^n$. After computing the limit, verify its distance to every stage center is less than that stage radius.

嵌套圆半径为 r^n、相继半径方向乘固定单位复数 u 时，圆心增量形成与 $(ru)^n$ 成比例的等比级数。求极限后，仍需检验它到每阶段圆心的距离小于该阶段半径。

Practice / 配套练习: AIME-2007-II-Q11 (WB p. 276), AIME-2017-II-Q12 (WB p. 94)

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 9 / 方法 9 [GE06-M09]

A rotational contact circle follows an axial similarity

旋转接触圆由轴截面相似确定

Procedure

1. Cut the rotational surfaces through their common axis.

2. Internal and external circle tangency give center distance $R-r$ and $R+r$, respectively.

3. The contact point lies on the center line; scale its distance from the axis by the corresponding radius ratio.

通用步骤

1. 作通过旋转曲面公共轴的截面。

2. 截圆内切、外切分别给出圆心距 $R-r, R+r$。

3. 切点在连心线上，其到轴距离按相应半径比缩放。

> Example. A radius-1 generating circle has center distance three from a rotation axis. It is externally tangent to an axial sphere of radius five. Find the contact-circle radius.

> 例题：半径一的母圆圆心距旋转轴为三，并与轴上圆心、半径五的球外切，求接触圆半径。

Worked solution

1. The two section centers are distance $5+1=6$ apart.

2. The contact point lies $5/6$ of the way from the sphere center to the generating-circle center.

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 9 / 方法 9 [GE06-M09] (continued / 续)

3. Its distance from the axis is $(5/6)3 = 5/2$, the contact-circle radius. The center placement exists because six exceeds three.

例题解析

1. 截面两圆心距为 $5 + 1 = 6$。

2. 切点从球心出发沿连心线走全长的 $5/6$。

3. 到轴距离为 $(5/6)3 = 5/2$；因六大于三，圆心轴向摆放确实存在。

> Use center distances in the section, not three-dimensional surface distances.

> 应使用截面中的圆心距，不能使用三维曲面距离。

Practice / 配套练习: AIME-2024-II-Q08 (WB p. 429)

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 10 / 方法 10 [GE06-M10]

An incidence quadratic combines two angle-bisector circles

过定点的二次方程联系两个角平分线圆

Procedure

1. Intersecting circles tangent to the same two lines occupy one common angular sector.

2. Parameterize their centers by (x, tx) and radii tx; a shared point gives one quadratic whose roots are the two center parameters.

3. Use Vieta to connect the radius product to t^2, choose the sector sign, and double the angle to recover the second tangent line.

通用步骤

1. 相交且同时与两直线相切的圆处于同一角域。

2. 把圆心写为 (x, tx)、半径为 tx；共同定点给出同一二次方程，两根即两圆心参数。

3. 用韦达定理把半径积联系到 t^2，筛选角域符号，再用倍角求另一切线。

> Example. Two circles through $(1, 1)$ have radius product $1/2$. Both are tangent to the x axis and a positive-slope line through the origin. Find that slope.

> 例题：两圆均过 $(1, 1)$，半径积为 $1/2$，并均与 x 轴及过原点的一条正斜率直线相切，求该斜率。

Worked solution

1. The common point above the axis places both circles above it. Their centers (x, tx) satisfy $(1 - x)^2 + (1 - tx)^2 = t^2x^2$, or $x^2 - 2(1 + t)x + 2 = 0$.

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 10 / 方法 10 [GE06-M10] (continued / 续)

2. Vieta gives $x_1x_2 = 2$, so the radius product is $2t^2 = 1/2$. The obtuse upper sector would have $t < -1$, impossible; hence $t = 1/2$.

3. The second tangent has slope $2t/(1 - t^2) = 4/3$. The center parameters are one and two, giving positive radii $1/2, 1$ and two genuinely intersecting circles.

例题解析

1. 共同点在轴上方，使两圆均位于上方。圆心 (x, tx) 满足 $(1 - x)^2 + (1 - tx)^2 = t^2x^2$，即 $x^2 - 2(1 + t)x + 2 = 0$。

2. 韦达给出 $x_1x_2 = 2$，故半径积为 $2t^2 = 1/2$。上方钝角域会有 $t < -1$，不可能，所以 $t = 1/2$。

3. 另一切线斜率为 $2t/(1 - t^2) = 4/3$。圆心参数为一与二，半径为 $1/2, 1$，确形成两个相交圆。

The sign of the angle-bisector slope carries the sector information.

角平分线斜率的符号包含角域信息。

Practice / 配套练习: AIME-2002-II-Q15 (WB p. 341)